

SAT UZBEKISTAN

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➤ Easy

1. If $ab > 0$, then which one of the following must be true?

- (A) $a/b > 0$
- (B) $a - b > 0$
- (C) $a + b > 0$
- (D) $b - a > 0$
- (E) $a + b < 0$

➤ Medium

2. If $a = x + 2y$, and $b = y + 2x$, and $3x + 7y > 7x + 3y$, then which one of the following is true?

- (I) $a > b$
- (II) $a = b$
- (III) $a < b$
- (A) I only
- (B) II only
- (C) III only
- (D) I and II only
- (E) II and III only

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3. If $x + z > y + z$, then which of the following must be true?

- (I) $x - z > y - z$
- (II) $xz > yz$
- (III) $x/z > y/z$
- (A) I only
- (B) II only
- (C) III only
- (D) I and II only
- (E) II and III only

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4. If $a = 3 + b$, which of the following is true?

- (I) $a > b + 2.5$
- (II) $a < b + 2.5$
- (III) $a > 2 + b$
- (A) I only
- (B) II only
- (C) III only
- (D) I and II only
- (E) I and III only

5. If $|x| + x = 4$, then which one of the following is odd?

- (A) $x^2 + 3x$
- (B) $x^2 + 3x + 2$
- (C) $x^2 + 4x$
- (D) $x^2 + 4x + 2$
- (E) $x^2 + 4x + 3$

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6. If $x < y < -1$, then which one of the following expressions is positive?

(A) $-x^2$
(B) y
(C) x^2y
(D) $\frac{x^2}{y^2}$
(E) $y - x^2$

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7. If $x + 3$ is positive, then which one of the following must be positive?

(A) $x - 3$
(B) $(x - 3)(x - 4)$
(C) $(x - 3)(x + 3)$
(D) $(x - 3)(x + 4)$
(E) $(x + 3)(x + 6)$

8. If $(x + 1)^2 - 2x > 2(x + 1) + 2$, then x cannot equal which one of the following?

(A) -5
(B) -3
(C) 0
(D) 3
(E) 5

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9. If $x^5 + x^2 < 0$, then which one of the following must be true?

(A) $x < -1$
(B) $x < 0$
(C) $x > 0$
(D) $x > 1$
(E) $x^4 < x^2$

10. In $\triangle PQR$, $PQ = x$, $QR = x + 3$, and $PR = y$. If $x = y + 3$, then which one of the following is true?

(A) $\angle P < \angle Q < \angle R$
(B) $\angle Q < \angle R < \angle P$
(C) $\angle R < \angle P < \angle Q$
(D) $\angle P < \angle R < \angle Q$
(E) $\angle Q < \angle P < \angle R$

11. If $5 < x < 10$ and $y = x + 5$, what is the greatest possible integer value of $x + y$?

(A) 18
(B) 20
(C) 23
(D) 24
(E) 25

12. If $x > 3y/5$, then which of the following inequalities must be true?

(I) $6y + 5x < 10x + 3y$
(II) $2y + 5x > 4x + 3y$
(III) $2y + 5x > 5x + 4y$

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

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13. If $(x - y)^3 > (x - y)^2$, then which one of the following must be true?

- (A) $x^3 < y^2$
- (B) $x^5 < y^4$
- (C) $x^3 > y^2$
- (D) $x^5 > y^4$
- (E) $x^3 > y^3$

14. If $x > 2$ and $x < 3$, then which of the following is positive?

- (I) $(x - 2)(x - 3)$
- (II) $(2 - x)(x - 3)$
- (III) $(2 - x)(3 - x)$

- (A) I only
- (B) II only
- (C) III only
- (D) I and II only
- (E) I and III only

15. If $x > y$ and $x < 0$, then which of the following must be true?

- (I) $\frac{1}{x} < \frac{1}{y}$
- (II) $\frac{1}{x-1} < \frac{1}{y-1}$
- (III) $\frac{1}{x+1} < \frac{1}{y+1}$

- (A) I only
- (B) II only
- (C) III only
- (D) I and II only
- (E) I and III only

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16. If $a^2 + 7a < 0$, then which one of the following could be the value of a ?

- (A) -3
- (B) 0
- (C) 1
- (D) 2
- (E) 3

➤ **Hard**

17. Three workers A, B, and C are hired for 4 days. The daily wages of the three workers are as follows:

A's first day wage is \$4.

Each day, his wage increases by 2 dollars.

B's first day wage is \$3.

Each day, his wage increases by 2 dollars.

C's first day wage is \$1.

Each day, his wage increases by the prime numbers 2, 3, and 5 in that order.

Which one of the following is true about the wages earned by A, B, and C in the first 4 days?

- (A) $A > B > C$
- (B) $C > B > A$
- (C) $A > C > B$
- (D) $B > A > C$
- (E) $C > A > B$

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18. If $x^2 - y^2 = 16$ and $x + y > x - y$, then which one of the following could $x - y$ equal?
- (A) 3
 - (B) 4
 - (C) 5
 - (D) 6
 - (E) 7
19. If it is true that $1/55 < x < 1/22$ and $1/33 < x < 1/11$, then which of the following numbers could x equal?
- I. $1/54$
 - II. $1/23$
 - III. $1/12$
- (A) I only
(B) II only
(C) III only
(D) I and II only
(E) II and III only

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➤ **Very Hard**

20. If $0 < x \leq 1$, then which one of the following is the maximum value of $(x - 1)^2 + x$?
- (A) -2
 - (B) -1
 - (C) 0
 - (D) 1
 - (E) 2

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